Predictive Modeling for Determination of Best Treatment Plan for Bell's Palsy to Accelerate Treatment Times

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Abstract

Bell's palsy is the most common cause of facial paralysis, affecting approximately 40,000 Americans annually. Despite its prevalence in the United States, the cause of Bell's palsy and the treatment plans for recovery still remain poorly understood. This study applies predictive modeling techniques to synthetic clinical trial data to evaluate the impact of age, sex, the time it took between the onset of symptoms and the start of treatment, Prednisolone, Acyclovir, and a combination of both on patient recovery by utilizing the changes in House-Brackmann (HB) score and whether full recovery was achieved, evaluated at 3 and 9 months. Feature selection analysis identified age and baseline HB score as the most consistent predictors across all outcomes, while Acyclovir showed no significant importance in predicting outcomes. Both regression and classification supervised learning approaches were explored, with regression models demonstrating poor predictive performance due to the discrete and categorical nature of the outcome variables. Classification models, particularly Random Forest, were found to be more appropriate, achieving high metrics across the target outcomes. Unsupervised clustering further supported Prednisolone as a meaningful factor in recovery. These findings highlight the importance of machine learning and predictive modeling in identifying key factors associated with the fastest recovery time for patients with Bell's palsy. Therefore, models will be able to assist clinicians to improve and optimize treatment strategy plans. Further studies should incorporate more clinical data to validate and improve models.

I. Introduction

Bell's palsy, the most common facial paralysis, is a neurological disorder that affects 40,000 people in the U.S. annually ("Bell's Palsy," n.d.). It primarily occurs when one of the nerves which controls muscles within the face (seventh cranial nerve) is injured or stops working. Symptoms include paralysis or weakness on one portion of the face, drooping eyebrow and mouth, drooling, and difficulty with closing one eyelid. In some cases, Bell's palsy can affect the whole face which can affect taste and ear sensation as well. Individuals most at risk of developing Bell's palsy are those within ages ranging from 15 to 45 (*Bell's Palsy | National Institute of Neurological Disorders and Stroke*, n.d.). While the specific cause of Bell's palsy is unknown, risk factors include pregnancy, preeclampsia, obesity, diabetes, and hypertension which can lead to inflammation in the seventh cranial nerve. To assess the severity of the condition, physicians use an internationally recognized six-tier facial nerve grading system known as the House-Brackmann (HB) scale. Severity of facial paralysis increases as HB grades increase from I (normal symmetrical function) - VI (total hemifacial paralysis) (Escalante et al.,

2021). The selection of treatment for Bell's palsy remains controversial and depends on the level of paralysis assessed by the physician. However, physicians most commonly prescribe corticosteroids (Prednisolone) or antiviral agents (Acyclovir) separately or in combination (Sullivan et al., 2007). The clinical data used assesses the effectiveness of these treatments for the early stages of Bell's palsy and their effect on the HB score and recovery time. The goal of the study is to use predictive modeling to determine whether patients who receive the combination of Prednisolone and Acyclovir will have a faster recovery time than those who receive one or neither of them.

II. Methodology

A. Data Collection

The dataset used in this study is from the online source, Kaggle, and consists of synthetic patient-level clinical trial data. The data was recreated based on the clinical trial data from a study in the New England Journal of Medicine (Sullivan et al., 2007). The original clinical trial followed a "double-blind, placebo-controlled, randomized factorial trial" design (Myrick, 2023). Patients with Bell's palsy were recruited within 72 hours after the onset of symptoms and were randomly assigned to receive Prednisolone, Acyclovir, both, or a placebo. Treatment was given for 10 days. The primary outcome was recovery of facial function, measured using the House-Brackmann scale, at 3 and 9 months post treatment.

The dataset contains 494 patients (52% male, 48% female, mean age 44.0 ± 16.4 years) with features including age (16-90), sex, baseline HB score, time between onset of symptoms and treatment, drug received (Prednisolone, Acyclovir, both, neither), and recovery time (3/9 months). Patients were approximately equally distributed across the four treatment groups, each representing roughly 25% of the cohort, as shown in Figure 1a. At 3-months, 358 patients (72.5%) achieved full recovery, increasing to 441 patients (89.3%) at 9-months, as shown in Figure 1b. Additional dataset statistics can be found in Appendix A.

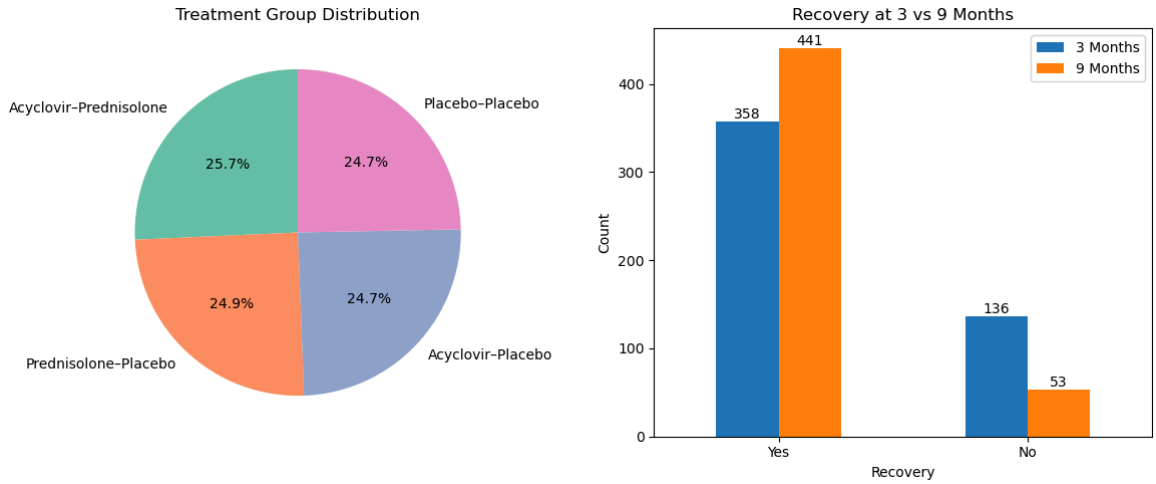


Figure 1. **(a)** Treatment group distribution for all 4 treatment groups (left). **(b)** Recovery comparison at 3 and 9 months (right).

B. Data Preprocessing

Prior to model training, the dataset was cleaned and prepared through a series of preprocessing steps. Categorical variables such as sex, drug received, and recovery were converted to binary and ordinal numerical format. The time between onset of symptoms and treatment was encoded as an ordinal variable representing increasing delay from symptom onset to treatment. The treatment group was removed because this information was represented by the drug received label. Any rows missing information was also dropped from the dataset.

The final feature set included, sex, age, baseline HB score, time between onset of symptoms and start of treatment, received Prednisolone, and received Acyclovir. All the features were then standardized (mean=0, standard deviation=1). Four target variables were used for analysis: the 3-month HB score, 3-month recovery, 9-month HB score, and 9-month recovery.

C. Data Splitting

The dataset was split using a comparative analysis of two strategies: Stratified K-Fold (SKF), with k=3, Cross-Validation and Leave-One-Out (LOO) Cross-Validation. These methods were selected because they maximized the use of the clinical sample size while ensuring that the training and testing subsets remained representative of the overall population distribution. Stratification was specifically implemented to preserve the distributions of HB grades across all folds, therefore mitigating the potential bias in class-imbalanced targets. The LOO approach served as a high-granularity benchmark to evaluate model generalizability across the entire patient cohort. By comparing the accuracy, precision, recall, and F1 scores of both splitting

methods, we identified the best strategy that yielded the highest model stability and lowest standard deviation for the subsequent stages of this project.

D. Feature Selection

To determine the most relevant features for our four target variables, we implemented a two-stage feature selection process. Given the clinical nature of the data and potential for class imbalance, all feature selection procedures were followed within a SKF Cross-Validation framework.

Due to the rarity of certain clinical outcomes, some minority classes contained fewer observations than the number of cross-validation folds. To address this, the HB scores were binned into three clinically distinct categories: Good (scores 1-2), Moderate (scores 3-4), and Severe (scores 5-6). This ensured class proportions were maintained as closely as possible and reported weighted metrics to account for this class imbalance.

1. Feature Selection Methods

We used six distinct feature selection methods, with the number of selected features fixed at k=3, to ensure that the chosen predictors were the most relevant and improved the model's performance. The baseline approach used all the features with no selection. Three wrapper methods were also applied: forward selection, which iteratively adds the most informative feature at each step; backwards selection, which iteratively removes the least informative feature, and bidirectional elimination, which combines the forward and backwards selection methods.

Additionally, two score comparison methods were used. The first was an ANOVA F-test to evaluate linear relationships between features and outcomes. The second was Mutual Information, which was used to measure the mutual dependence between two variables.

2. Stability Scoring

To ensure that the selected features were generalizable and not overfit to specific data splits, we calculated a Methodological Consensus Stability Score for each candidate feature. Following the frequency-based stability framework in Nogueira et al. (2017), stability was used as a proportion of times a given feature was selected across all cross-validation folds and feature selection methods. While Nogueira et al. (2017) defines it with respect to a single algorithm, we extended this principle across our identified five feature selection methods, in addition to the stratified cross-validation folds.

Features were prioritized for the final predictive models based on this cumulative stability. By integrating selection within the SKF loop, we ensured that the resulting feature sets

were robust against variations in the training data, prioritizing variables that demonstrated consistency across multiple clinical subgroups.

E. Model Selection

1. Supervised Learning

For the supervised learning phase, we initially attempted to do a regression-based approach to predict HB score and recovery time at 3 and 9 months. Four regression models were evaluated: Linear Regression, Decision Tree Regression (DTR), Random Forest Regression (RFR), and Support Vector Regression (SVR). Feature variables were standardized such that each feature had a mean of 0 and a standard deviation of 1. The scaler was then fitted on the training and testing data. However, due to the limited performance of the regression-based approach and the categorical nature of both outcome variables, classification methods were deemed more appropriate.

As a result, we then evaluated three different classification models to identify the optimal balance between clinical interpretability and predictive power. Logistic Regression was implemented as the primary linear baseline, chosen for its transparency and established utility in clinical applications. To account for potential complexities in the data that linear models might overlook, we also employed Random Forest Classification (RFC) and Support Vector Machine (SVM) classifiers. The RFC algorithm was used to capture high-order, non-linear interactions through an ensemble of decision trees, while SVM was chosen for its ability to identify an optimal separation hyperplane within a high-dimensional feature space. Following the identification of the best model that fits the data, via GridsearchCV() we determined the optimal hyperparameters using accuracy to tune the model for the best metrics.

To address the class distribution challenges, all the models utilized a balanced class-weighting strategy, ensuring that the minority recovery outcomes were sufficiently prioritized during the training process.

2. Unsupervised Learning

Unsupervised learning models were applied to explore the latent patterns without the constraint of predefined labels. To minimize noise and eliminate multicollinearity among the clinical variables, Principal Component Analysis (PCA) was performed as a preliminary dimensionality reduction step. The analysis was restricted to a maximum of three principal components (PCs) to allow easy visualization of the data while capturing the majority of the variance. This allowed for the transformation of the feature set into a series of orthogonal components that preserved the maximum variance of the original data. Following PCA, the data

was analyzed using two clustering methods: K-Means Clustering with K-Means++ initialization and Hierarchical Clustering. K-Means was employed to split the data into distinct, non-overlapping groups based on Euclidean distance, while Hierarchical Clustering was used to reveal the nested relationships and subgroups through a dendrogram-based approach. This comparison of methods provided a comprehensive view of the underlying data architecture, helping to identify the distinct patient recovery groups.

F. Evaluation Metrics

Model performance was assessed using mean squared error (MSE) and coefficient of determination (R^2) for regression models and accuracy, weighted precision, weighted recall, and weighted F1-score for all subsequent evaluations. Given the clinical nature of the dataset and the inherent imbalance in recovery outcomes, weighted averaging was used for all multiclass and binary targets. This approach accounts for class frequency by calculating metrics for each label and finding their average, weighted by the number of true instances for each label. These metrics were applied consistently across all four target variables to ensure standardized comparison. To evaluate the stability of these predictions, results were reported as the mean value across cross-validation folds, in addition to the standard deviation.

III. Results

A. Dataset Split Selection Results

Table 1. Comparison of SKF vs. LOO Dataset Split

3-Month HB Score			3-Month Full Recovery		
Metric	**SKF Mean**	**LOO Mean**	**Metric**	**SKF Mean**	**LOO Mean**
Accuracy	0.507	0.527	**Accuracy**	0.657	0.675
Precision	0.663	0.527	**Precision**	0.712	0.675
Recall	0.507	0.527	**Recall**	0.657	0.675
F1 Score	0.570	0.527	**F1 Score**	0.672	0.675
9-Month HB Score			**9-Month Full Recovery**		
Metric	**SKF Mean**	**LOO Mean**	**Metric**	**SKF Mean**	**LOO Mean**
Accuracy	0.564	0.544	**Accuracy**	0.675	0.659
Precision	0.836	0.544	**Precision**	0.859	0.659
Recall	0.564	0.544	**Recall**	0.675	0.659
F1 Score	0.661	0.544	**F1 Score**	0.735	0.659

Table 1 compares SKF and LOO cross-validation across all four target outcomes. SKF consistently outperformed LOO in precision and F1 score, most notably in 9-month HB score precision (0.836 vs 0.544), and was therefore selected as the primary splitting strategy for all subsequent modeling.

B. Feature Selection Results

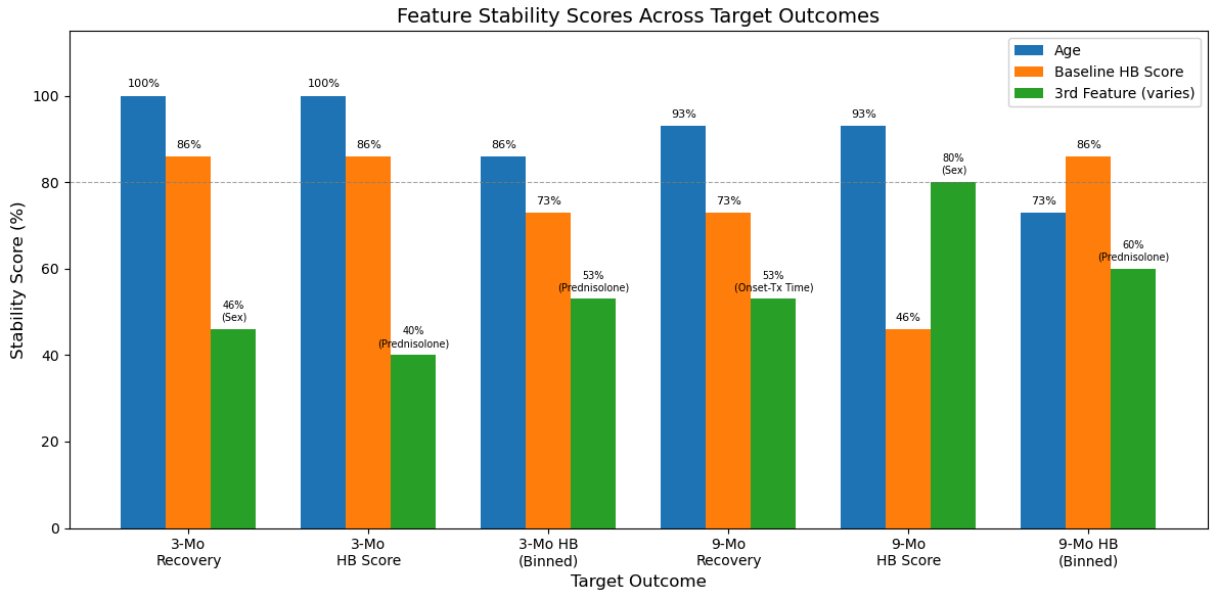


Figure 2. Feature selection of top 3 predictors across target outcomes with the associated stability score.

Figure 2 shows the stability scores for the top three features across all six targets. Age and baseline score were the most consistently selected predictors across all outcomes. The third feature varied by target, with Prednisolone and sex appearing most frequently, while Acyclovir did not appear among the top three features for any target. The tabular version of these results can be found in Table X of Appendix B.

C. Model Selection Results

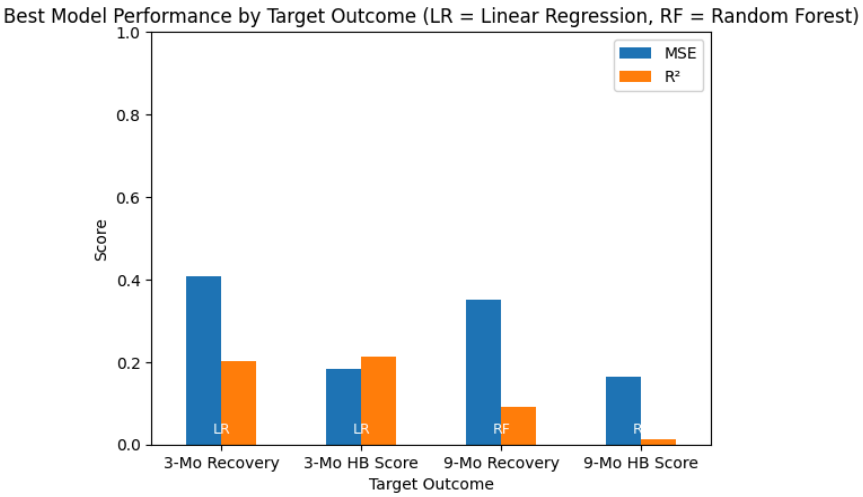


Figure 3. Comparison of best regression model, chosen from Linear Regression, RFR, and SVR, across 3-month recovery and HB score and 9-month recovery and HB score.

Figure 3 compares the best model performance across the four targets. The 3-months targets perform better with Linear Regression and the 9-month targets perform better with RFR. Overall, the performance of the models performed poorly across all targets.

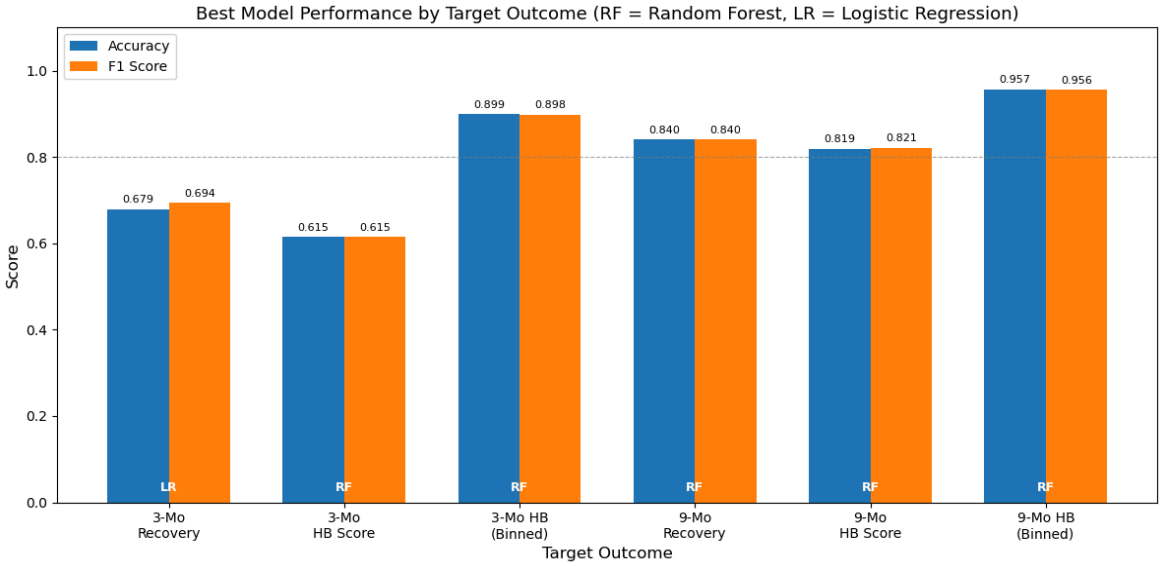


Figure 4. Comparison of best classification model, chosen from Logistic Regression, RFC, and SVM, across all target variables.

Figure 4 compares the best model performance across all six targets. RFC outperformed Logistic Regression and SVM in five of six targets, with strongest results on the binned score outcomes (F1: 0.898 at 3-months, 0.956 at 9-months). Logistic Regression was the best model

only for 3-month full recovery, while the unbinned 3-month HB score remained the most difficult target across all models (best accuracy 0.615). The tabular version of these results can be found in Table X of Appendix B.

1. Random Forest Tuning Results

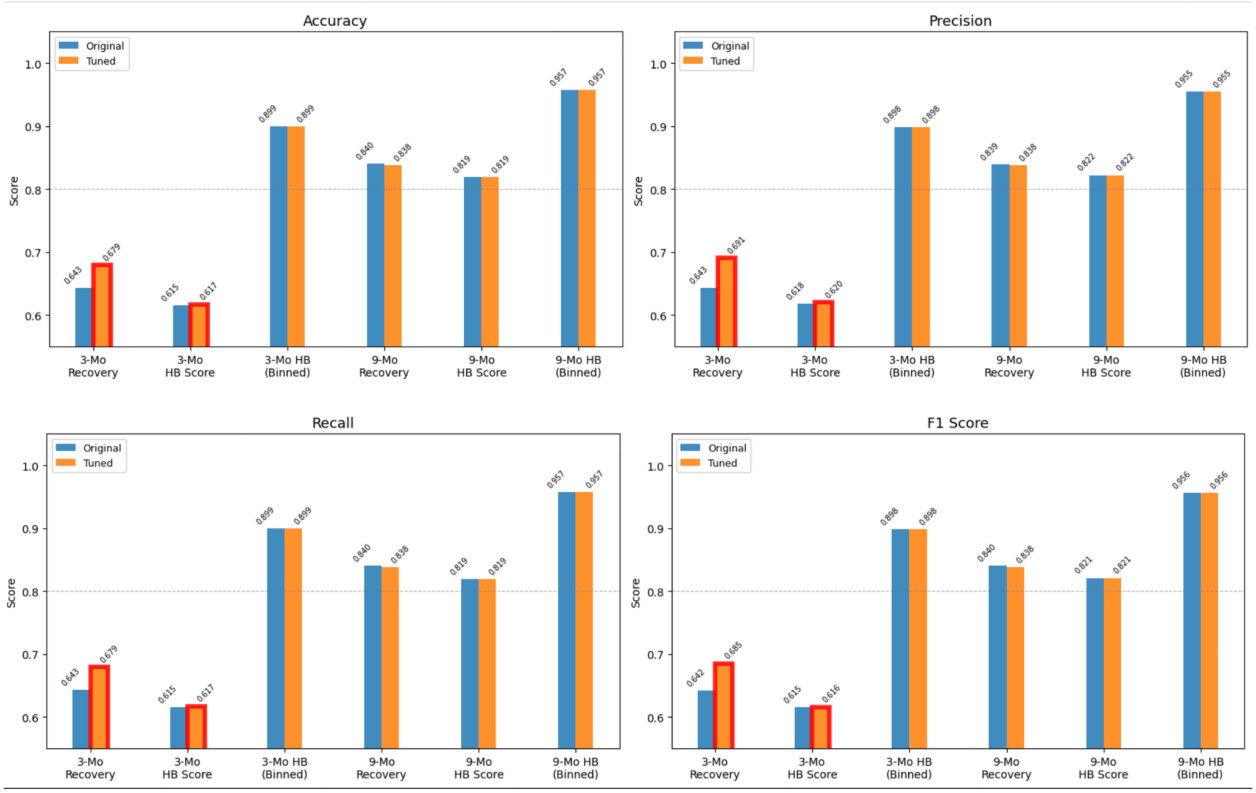


Figure 5. Comparison of original and tuned hyperparameter performance across accuracy, precision, recall, and F1 score using RFC model. Bars highlighted red indicated improvement as a result of tuning.

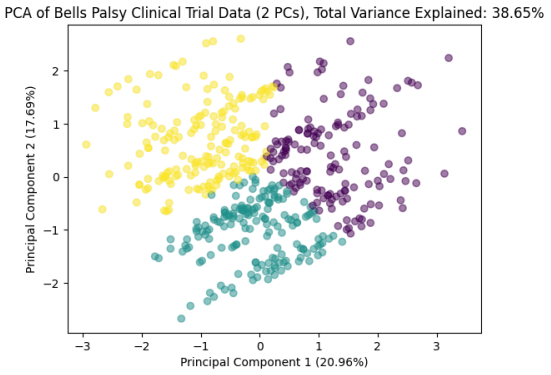
Figure 5 compares RFC performance before and after hyperparameter tuning. The most notable improvement was in 3-month full recovery, where F1 score increased from 0.642 to 0.685. Performance on all other targets remained largely unchanged, indicating they were already well-optimized prior to tuning.The tabular version of these results can be found in Table X of Appendix B.

D. Unsupervised Learning Results

Table 2. Clustering Profiles Comparing K-Means and Hierarchical Methods using 3 PCs						
Feature (Yes %) /Cluster	**K-Means Clustering Profile (3 PCs)**			**Hierarchical Clustering Profile (3 PCs)**		
	Cluster 0	**Cluster 1**	**Cluster 2**	**Cluster 0**	**Cluster 1**	**Cluster 2**
Avg. Age	54.9	38.2	42.8	50.6	41.3	40.0
Avg. Baseline HB	4.56	3.23	3.30	4.10	3.30	3.40
Prednisolone Use	29.7%	36.5%	91.8%	25.8%	94.6%	41.4%
Acyclovir Use	41.1%	56.1%	52.7%	38.3%	56.4%	61.4%
Sex (M/F)	29.7/70.3 %	72.5/27.5 %	48.6/51.4 %	24.2/75.8 %	46.9/53.1 %	97.1/2.9 %
3-Mo Recovery	55.7%	78.8%	82.9%	64.0%	83.7%	74.3%
9-Mo Recovery	81.6%	92.1%	94.5%	84.4%	95.2%	90.7%
3-Mo Avg. HB Score	1.59	1.24	1.19	1.50	1.20	1.30
9-Mo Avg. HB Score	1.28	1.09	1.06	1.20	1.00	1.10

Table 2 summarizes the cluster profiles for both K-Means and Hierarchical clustering using 3 PCs. Across both methods, clusters were primarily differentiated by age, baseline HB severity, and Prednisolone use. In both methods, the cluster with the highest Prednisolone use (K-Means Cluster 2: 91.8%, Hierarchical Cluster 1: 94.6%) consistently achieved the best 3-month and 9-month recovery rates and lowest average HB scores at both timepoints. Conversely, the cluster with the oldest patients and highest baseline severity (K-Means Cluster 0: age 54.9, HB 4.56; Hierarchical Cluster 0: age 50.6, HB 4.1) showed the weakest recovery outcomes in both methods. Acyclovir use did not show a consistent relationship with recovery outcomes across any cluster in either method. The full table can be found in Appendix B.

1. K-Means Clustering



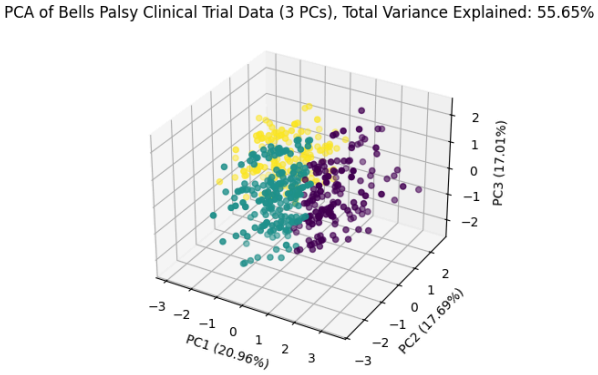


Figure 6. (a) PCA of clinical trial data using 2 PCs (left). **(b)** PCA of clinical trial data using 3 PCs (right).

Figure 6 shows the scatter plots for K-Means clustering using 2 (Figure 6a) and 3 (Figure 6b) PCs respectively. The 2 PC model explained 38.65% of the total variance, while the 3 PC model improved this to 55.65%. In both models, no clear visual separation between clusters is apparent, suggesting the dataset does not contain strongly distinct patient groupings in the reduced feature space.

2. Hierarchical Clustering

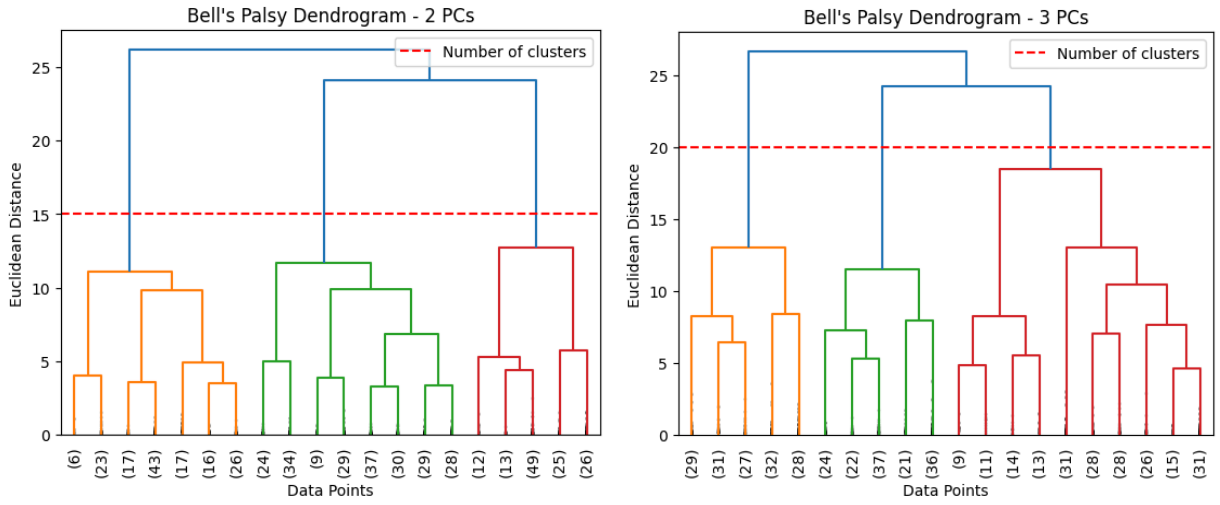


Figure 7. (a) Hierarchical clustering dendrogram using 2 PCs (left). **(b)** Hierarchical clustering dendrogram using 3 PCs (right).

Figure 7 shows the hierarchical clustering dendrograms for the 2PC (Figure 7a) and 3PC (Figure 7b) models respectively. In both cases, the dashed cut off line indicates three clusters as the optimal number, consistent with the K-Means results. The similar cluster structure across both methods and both PC configurations further supports the stability of the three identified patient groupings.

IV. Discussion

A. Methodology Decisions

To determine the optimal strategy for splitting the dataset, SKF cross-validation and LOO cross-validation were compared across all four target outcomes. Overall, neither method demonstrated a consistently superior performance across all metrics, with differences being marginal depending on the metric. For the 3-month targets, LOO had marginally higher accuracy and recall for both HB scores (0.527 vs. 0.507) and full recovery (0.675 vs. 0.657).

However, SKF outperformed LOO in precision and F1 score for both targets, most notably in HB score precision (0.633 vs 0.527), suggesting that SKF produces more balanced predictions across classes despite lower raw accuracy. For the 9-month targets, SKF demonstrated a stronger performance across all metrics. The 9-month HB score showed the largest difference with SKF precision reaching 0.836 compared to 0.544 for LOO. The consistently higher precision and F1 scores under SKF are meaningful in a clinical context where distinguishing between recovery outcomes accurately across all classes is more informative than simple accuracy alone. SKF was therefore selected as the primary dataset splitting strategy for subsequent modeling, as it provided greater stability, better class representation, and more reliable weighted metrics.

Feature selection was performed across six methods within a Stratified 3-Fold cross-validation framework, with stability scores calculated to identify the most consistently selected features across folds and methods. Age and baseline HB score were the most stable and clinically relevant predictors across nearly all targets. Age achieved perfect or near perfect stability for the 3-month full recovery (100%), 3-month HB score (100%), 9-month full recovery and HB score (93%), while the baseline HB score ranged from 46% to 86% across all outcomes. The consistent selection of these two features aligns with the established clinical understanding that age and presenting severity are among the strongest prediction indicators for Bell's palsy. The third feature varied by target. For the 3-month outcomes, Sex (46%) and Received Prednisolone (40%-53%) appeared as third features with relatively low stability, suggesting modest contributions at the earlier time points. For the 9-month outcomes, Time Between Onset and Treatment (53%), Sex (80%), and Received Prednisolone (60%) were the third features with notably higher stability scores, suggesting these features became more predictively relevant over long recovery periods. Acyclovir did not appear among the top three features for any target, indicating its independent predictive contribution was consistently outweighed by other variables.

B. Supervised Learning

A regression-based approach was initially conducted to predict the HB score and full recovery of patients with Bell's palsy in 3 months and 9 months. All four regression models demonstrated very low predictive power, with R^2 values near zero across most outcomes. For the 3-month score, the highest R^2 value and lowest MSE was found to be the linear regression model with a value of 0.201 and 0.409, respectively. Similarly, the 3-month full recovery had a maximum R^2 value of .182 and a minimum MSE value of 0.213, both from the linear regression

model. For the 9-month score and 9-month full recovery, both the maximum R^2 value and minimum MSE value came from random forest regression with a value of $R^2 = 0.09$ and MSE = 0.350 for the 9-month score and a value of $R^2 = 0.013$ and MSE = 0.165 for the 9-month full recovery. Other R^2 values were either lower or even negative, signifying very poor generalization and failure to predict patient outcomes. Visualization of the predicted vs actual values for the 3-month score further highlights the failures of using a regression model. From Figure 12 in Appendix B, it can be seen that the model is far from the perfectly predicted line (dashed line) and is more likely to underpredict values. Additionally, from Figure 12 in Appendix B, it can be seen that the predicted values are still far from the perfectly predicted line. Additionally, in both plots, there are only vertical clusters instead of a continuous regression line, emphasizing that regression is not appropriate for this model.

All four regression models failed to predict the outcome variables accurately because of the nature of the HB score as well as the binary non-continuous outcome for predicting full recovery. The HB score is a range from 1-6 and while it seems continuous, it is actually discrete because each score represents a single, specific category and intermediate values (such as 1.3) do not represent anything within this scale. Therefore, treating this problem as a regression model is incorrect since it assumes continuity between each value. Similarly, a regression model for predicting full recovery in 3 months and 9 months also assumes continuity between 0 (no) and 1 (yes), which isn't true because intermediate values (like 0.5) have no meaning. Ultimately, a classification-based approach was more appropriate for modeling and motivated the transition to a classification-based approach.

Following the failure of regression-based approaches, the three classification models were evaluated: Logistic Regression, RFC, and SVM. Random Forest was the best-performing model across the majority of outcomes, as shown in Figure 4, reflecting its ability to capture non-linear interactions between features that linear models cannot. The binned HB score targets yielded the strongest results across both timepoints (F1: 0.898 at 3-months, 0.956 at 9-months), validating the decision to aggregate HB grades to address class imbalance. Logistic Regression was the best model only for 3-month full recovery (accuracy 0.679, F1 0.694), suggesting that the relationship between the selected features and short-term binary recovery is sufficiently linear to be captured by the simpler model. The relatively modest performance on the unbinned 3-month HB score (best accuracy 0.615) highlights the difficulty of predicting fine-grained severity at the earlier recovery timepoint, where outcomes are more variable.

Following model selection, hyperparameter tuning was applied to the RFC classifier. The most meaningful improvement was observed for 3-month full recovery, where accuracy increased from 0.643 to 0.679 and F1 from 0.642 to 0.685. For the remaining targets, performance was largely unchanged, indicating they were already well-optimized prior to tuning. The narrow gap between precision, recall, and F1 across all outcomes following tuning indicates that the balanced class-weighting strategy was effective in preventing the model from disproportionately favoring majority classes. Overall, the tuned RFC model represents a clinically viable tool for Bell's palsy outcome prediction, particularly for 9-month prediction and binned severity classification.

C. Unsupervised Learning

While the data is labeled, PCA and K-Means clustering was also applied to the dataset to explore any underlying patterns without the constraint of predefined labels. These unsupervised techniques were used to identify any potential patterns or separability among patients that may not be as obvious when looking at labeled outcomes. 2 and 3 PCs were evaluated, explaining 38.65% and 55.65% of the total variance, respectively. No additional components were added because it would be too challenging to visualize the clusters.

From the plots using K-Means clustering (Figure 6), it can be seen that there was no significant separation between each cluster for both the 2D and the 3D model. Additionally, looking at the tables for each cluster, no significant or distinct patterns from the features can be found that affect the HB score or the full recovery of a patient. For example, for the 2D clustering, cluster 0 had the lowest number of patients that used Prednisolone which equated to less people recovering in 3 months while cluster 2 had the highest number which equated to more people recovering in 3 months. However, cluster 0 also had a higher average baseline score than cluster 2, indicating that Prednisolone might not be the only factor and as a result, no meaningful correlation was found. While the data does suggest that patient recovery improves over time, this is likely due to the natural healing process of the body rather than any distinctive features that are involved. Ultimately, the tables and plots of the clusters signify that the dataset does not have any clear groupings and supervised learning using classification techniques are more appropriate to predict outcome variables.

Hierarchical clustering was similarly performed to identify patient subgroups or patterns that may have not been apparent within the dataset. The specific approach used was agglomerative (bottom-up) which repeatedly groups smaller clusters until the largest cluster is

identified. To determine the number of clusters, the agglomerative clustering dendrogram indicated that three clusters were identified to determine the labeling for the groups (Figure 7).

As previously mentioned, due to the high dimensionality of the dataset the clustering was initially done with 2 PCs to reduce the number of features while simultaneously preserving the variability in the data. However, the components only represented ~38% of the variability so the number of PCs was increased to 3 to capture 55.56% of the data. Through comparison of both two and three PCs the stability of patient groupings and the relative importance of clinical features in determining recovery were assessed. Across both the 2 and 3 PC PCA models, clustering revealed distinct patient subgroups primarily based on age, baseline severity, and treatment distribution. In the 2 PC model, Cluster 1 was characterized by older patients (average age 56.6 years) and a higher baseline severity (average score 4.6) whereas Clusters 0 and 2 consisted of slightly younger patients (~38-42 years) with lower baseline severity (3.3). This separation could suggest that age and initial disease severity are significant factors in variation of groups.

Treatment distribution varied substantially across clusters. In both PCA approaches, certain clusters were heavily associated with Prednisolone use, while others had low exposure. For example, in the three-component model, Cluster 1 has 94.6% Prednisolone usage, whereas Cluster 0 only had 25.8%. This then translates into an 83.7% full recovery in 3-months, 95.2% full recovery in nine months, and lower average scores at each time point (3-Months = 1.2, 9-Months 1.0). Cluster 0 showed lower recovery scores and a higher average HB score at each time point. Acyclovir usage showed less consistent positive outcomes across clusters, indicating its role as a treatment may not be as effective as Prednisolone.

V. Conclusion

Through the exploration of supervised and unsupervised machine learning methods including regression, classification, and clustering we were able to identify the best approaches to identifying the best approach to treating patients with Bell's palsy. The first models evaluated were regression models such as Linear Regression, DTR, RFR, and SVR. It was determined that the regression models were not suitable for this dataset. Using classification models, it was determined that RFC was best suited for prediction modeling for the Bell's palsy dataset with an accuracy and F1 score of 0.838 for 9-month full recovery and 0.819/0.821 for the 9-month HB score. To determine the implications for treatment effectiveness, K-Means clustering and hierarchical clustering were performed. The results suggest that clusters with high Prednisolone use were associated with improved recovery outcomes supporting it as an effective treatment

for Bell's palsy. In contrast, clusters with higher Acyclovir use did not show a clear consistent relationship with improved recovery, aligning with clinical evidence which suggest antiviral therapy provides limited additional benefit (Allen, 2004). Overall, this study demonstrates the value of machine learning in supporting clinical decision-making for Bell's palsy treatment. Future work should focus on integrating these predictive models into personalized medicine frameworks, where treatment recommendations can be tailored to individual patient factors such as age, sex, baseline HB score, symptom severity, and treatment timing. By leveraging data-driven tools, clinicians may be better equipped to make informed treatment decisions and optimize recovery outcomes for patients.

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Appendix A: Dataset Statistics

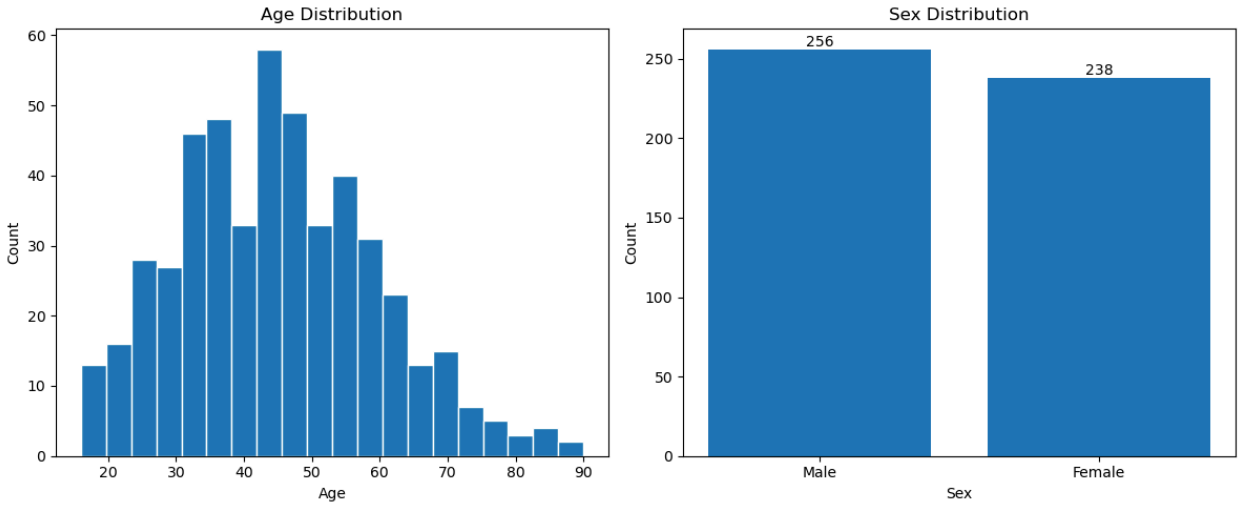


Figure 8. (a) Age distribution of raw dataset (left). **(b)** Distribution of raw dataset between male and female (right).

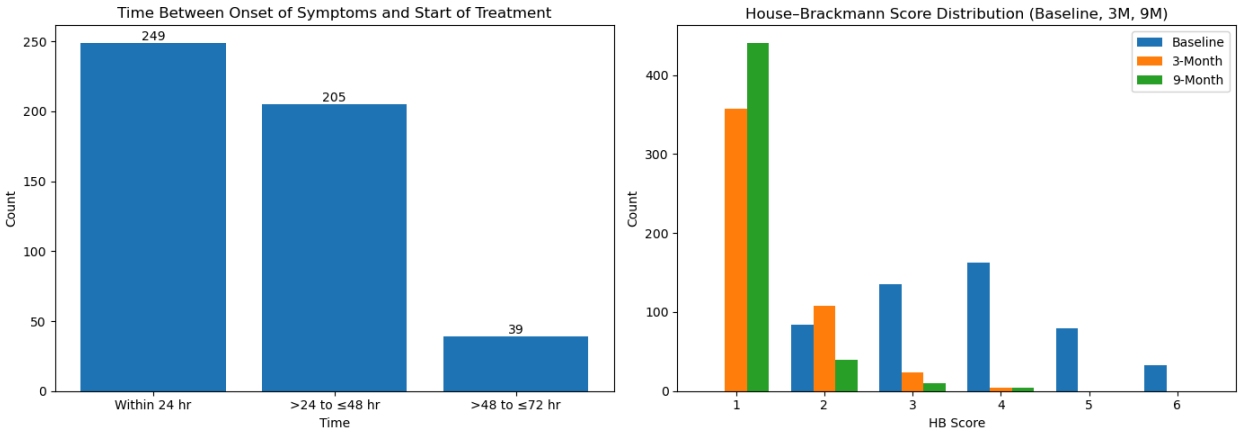


Figure 9. (a) Distribution of time between onset of symptoms and start of treatment grouped into the three time periods provided in the raw dataset (left). **(b)** Distribution of HB scores across the baseline, 3-months, and 9-months (right).

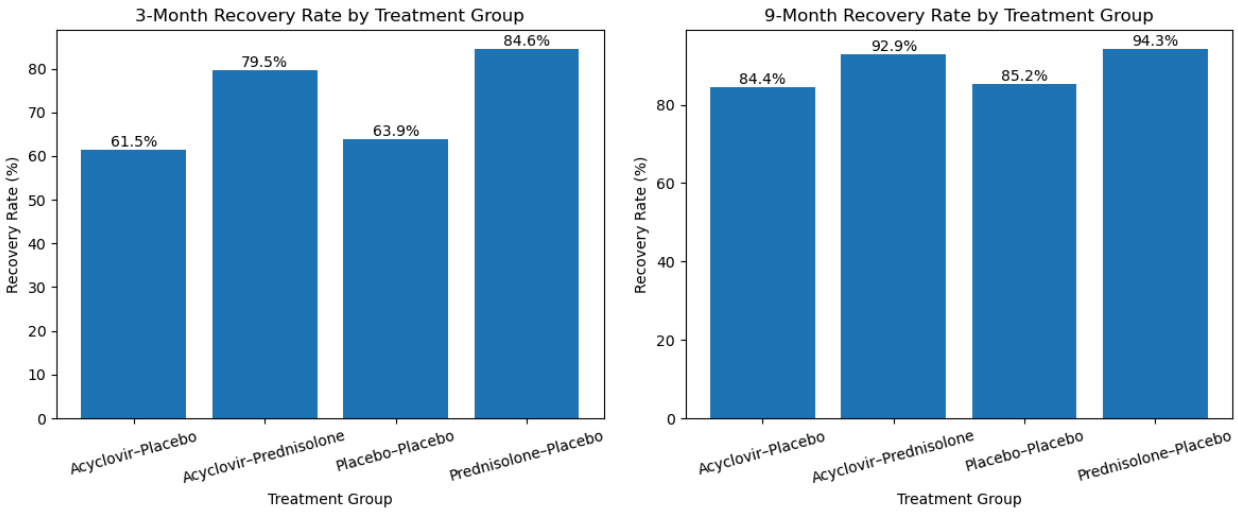


Figure 10. (a) Distribution of the recovery rate at 3-months for each of the four treatment groups (left). **(b)** Distribution of the recovery rate at 9-months for each of the four treatment groups (right).

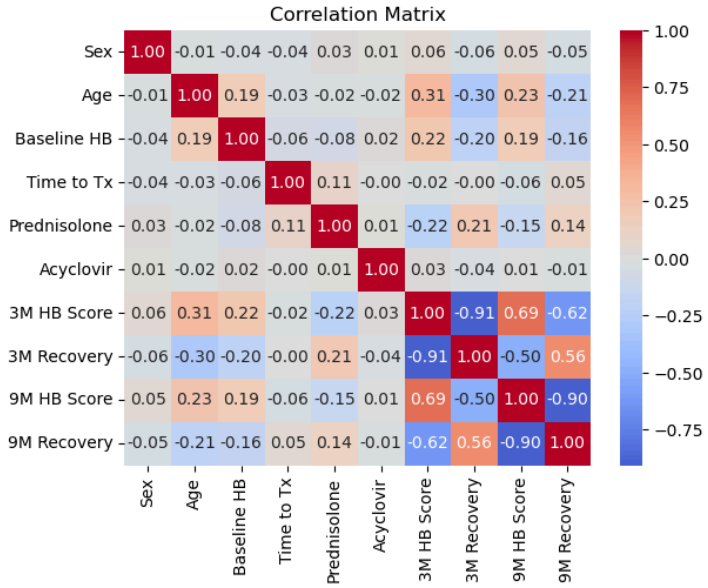


Figure 11. Confusion matrix of features included in this study.

Appendix B: Raw Results

Table 3. Top 3 Features for Each Target Variable with Associated Stability Score			
Target Outcomes	**Feature 1 (stability)**	**Feature 2 (stability)**	**Feature 3 (stability)**
3-Month Full Recovery	Age (100%)	Baseline Score on HB Scale (86%)	Sex (46%)
3-Month HB Score	Age (100%)	Baseline Score on HB Scale (86%)	Received Prednisolone (40%)
3-Month HB Score (Binned)	Age (86%)	Baseline Score on HB Scale (73%)	Received Prednisolone (53%)
9-Month Full Recovery	Age (93%)	Baseline Score on HB Scale (73%)	Time Between Onset and Treatment (53%)
9-Month HB Score	Age (93%)	Sex (80%)	Baseline Score on HB Scale (46%)
9-Month HB Score (Binned)	Baseline Score on HB Scale (86%)	Age (73%)	Received Prednisolone (60%)

Model Selection Results - Regression

Table 4. Results from Model Selection Comparing Linear Regression,DTR, RFR, and SVR			
Target Outcome	**Best Model**	**MSE**	**R2-Score**
3-Months Full Recovery	Linear Regression	0.182	0.213
3-Months HB Score	Linear Regression	0.409	0.201
9-Months Full Recovery	Random Forest Regression	0.164	0.019
9-Months HB Score	Random Forest Regression	0.354	0.081

Table 5. Classification Model Selection Comparing Logistic Regression, RFC, SVM			
Target Outcome	**Best Model**	**Accuracy**	**F1 Score**
3-Month Full Recovery	Logistic Regression	0.679	0.694
3-Month HB Score	**Random Forest**	0.615	0.615
3-MonthHB Score (Binned)	**Random Forest**	0.899	0.898
9-Month Full Recovery	**Random Forest**	0.84	0.84
9-Month HB Score	**Random Forest**	0.819	0.821

Table 5. Classification Model Selection Comparing Logistic Regression, RFC, SVM			
9-Month HB Score (Binned)	**Random Forest**	0.957	0.956

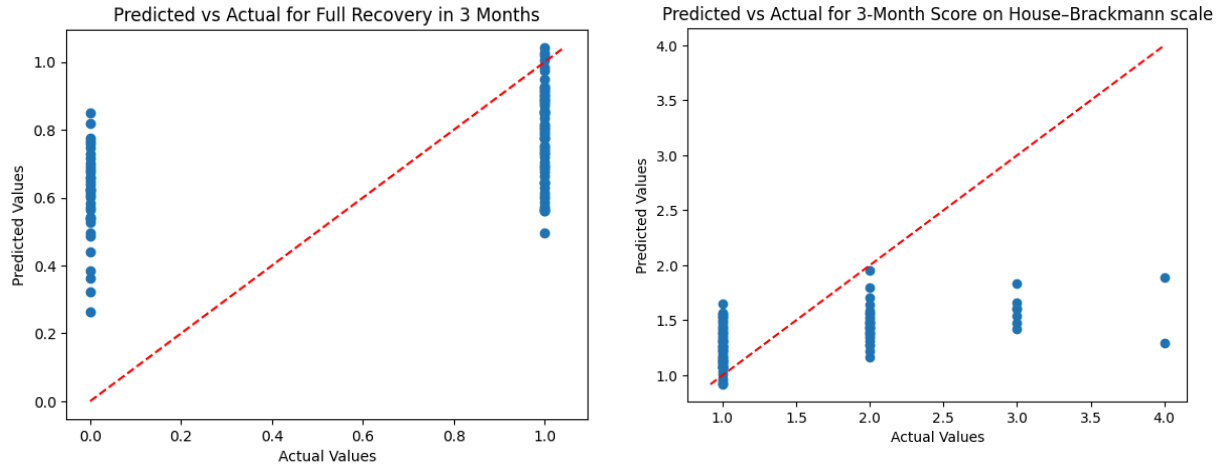


Figure 12. Predicted vs. Actual comparison for 3-month full recovery and HB scale results.

Table 6. Random Forest Model Metrics After Tuning Hyperparameters								
Target Outcome	**Accuracy**		**Precision**		**Recall**		**F1 Score**	
	Tuned	**Original**	**Tuned**	**Original**	**Tuned**	**Original**	**Tuned**	**Original**
3-Month Full Recovery	0.679	0.643	0.691	0.643	0.679	0.643	0.685	0.642
3-Month HB Score	0.617	0.615	0.620	0.618	0.617	0.615	0.616	0.615
3-Month HB Score (Binned)	0.899	0.899	0.898	0.898	0.899	0.899	0.898	0.898
9-Month Full Recovery	0.838	0.840	0.838	0.839	0.838	0.840	0.838	0.840
9-Month HB Score	0.819	0.819	0.822	0.822	0.819	0.819	0.821	0.821
9-Month HB Score (Binned)	0.957	0.957	0.955	0.955	0.957	0.957	0.956	0.956

Table 7. PCA and K-Means Clustering using 2 PCs								
	Prednisolone		Acyclovir		Sex		Average Baseline Score	Average Age
	Yes (%)	No (%)	Yes (%)	No (%)	M	F		
Cluster 0	36.7	63.3	42.9	57.1	34.7	65.3	4.64	57.1
Cluster 1	29.3	70.7	57.5	42.5	77.6	22.4	3.34	38.5
Cluster 2	84.3	15.7	49.4	50.6	40.1	59.9	3.20	40.85

Table 8. PCA and K-Means Clustering using 2 PCs - Results						
	3-Month Full Recovery		3-Month Average Score	9-Month Full Recovery		9-Month Average Score
	Yes (%)	No (%)		Yes (%)	No (%)	
Cluster 0	55.1	44.9	1.60	81.6	18.4	1.29
Cluster 1	77.0	23.0	1.27	90.2	9.80	1.11
Cluster 2	83.1	16.9	1.19	95.3	4.70	1.05

Table 9. PCA and K-Means Clustering using 3 PCs								
	Prednisolone		Acyclovir		Sex		Average Baseline Score	Average Age
	Yes (%)	No (%)	Yes (%)	No (%)	M	F		
Cluster 0	29.7	70.3	41.1	58.9	29.7	70.3	4.56	54.93
Cluster 1	36.5	63.5	56.1	43.9	72.5	27.5	3.23	38.22
Cluster 2	91.8	8.20	52.7	47.3	48.6	51.4	3.30	42.8

Table 10. PCA and K-Means Clustering using 3 PCs - Results						
	3-Month Full Recovery (Yes %)		3-Month Average Score	9-Month Full Recovery		9-Month Average Score
	Yes (%)	No (%)		Yes (%)	No (%)	
Cluster 0	55.7	44.3	1.59	81.6	18.4	1.28
Cluster 1	78.8	21.2	1.24	92.1	7.90	1.09
Cluster 2	82.9	17.1	1.19	94.5	5.50	1.06

Table 11. PCA and Hierarchical Clustering using 2 PCs								
	Prednisolone		Acyclovir		Sex		Average Baseline Score	Average Age (years)
	Yes (%)	No (%)	Yes (%)	No (%)	M	F		
Cluster 0	90.4	9.6	48	52	42%	58%	3.3	42.2
Cluster 1	34.4	65.6	45	55	37%	63%	4.6	56.6
Cluster 2	39.1	60.9	55	45	67%	33%	3.3	38.5

Table 12. PCA and Hierarchical Clustering using 2 PCs - Results						
	3-Month Full Recovery		3-Month Average Score	9-Month Full Recovery		9-Month Average Score
	Yes (%)	No (%)		Yes (%)	No (%)	
Cluster 0	84.0	16.0	1.18	95.2	4.8	1.1
Cluster 1	54.7	45.3	1.60	81.8	18.2	1.2
Cluster 2	78.1	21.9	1.3	91.4	8.6	1.1

Table 13. PCA and Hierarchical Clustering using 3 PCs								
	Prednisolone		Acyclovir		Sex		Average Baseline Score	Average Age (years)
	Yes (%)	No (%)	Yes (%)	No (%)	M (%)	F (%)		
Cluster 0	25.8	74.2	38.3	61.7	24.2	75.8	4.1	50.6
Cluster 1	94.6	5.4	56.4	43.6	46.9	53.1	3.3	41.3
Cluster 2	41.4	58.6	61.4	38.6	97.1	2.9	3.4	40.0

Table 14. PCA and Hierarchical Clustering using 3 PCs - Results						
	3-Month Full Recovery		3-Month Average Score	9-Month Full Recovery		9-Month Average Score
	Yes (%)	No (%)		Yes (%)	No (%)	
Cluster 0	64.0	36.0	1.5	84.4	15.6	1.2
Cluster 1	83.7	16.3	1.2	95.2	4.8	1.0
Cluster 2	74.3	25.7	1.3	90.7	9.3	1.1